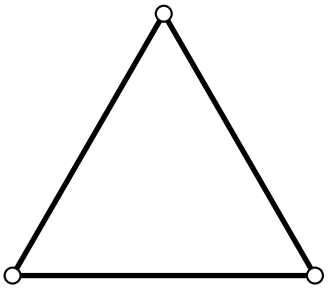
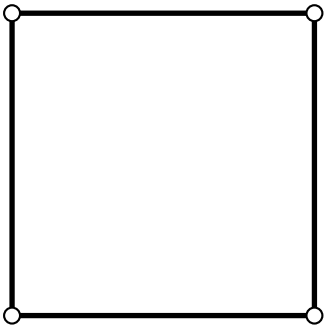
Keep the frames flat. Bars stay straight and keep their length; joints may turn. Sliding or turning the whole frame does not change its shape.

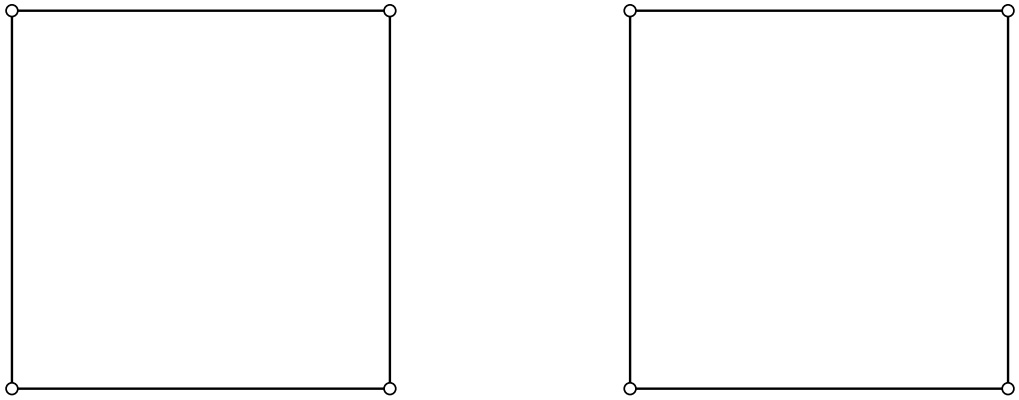
Problem 1: Can you change the shape of the triangle and the square? Draw the shapes you can make.





Problem 2: Make the square hold its shape with the fewest diagonal bars. Draw where you put the bars.

A diagonal bar joins opposite corner pins. It is called a brace.

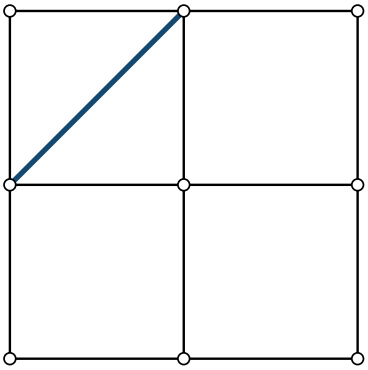


Problem 3: Remove the diagonal and make shapes with the four side bars. Which shapes can you fit that same diagonal into?

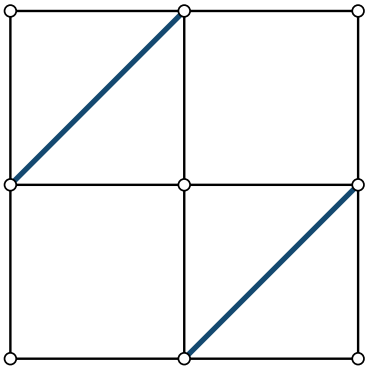
Keep the frames flat. Moving or turning the whole frame does not change its shape. In a grid, use at most one diagonal brace in each cell.

Problem 4: Build these designs on your 2-by-2 frame. Circle each design that holds its shape. For each frame that can change shape, draw a changed shape.

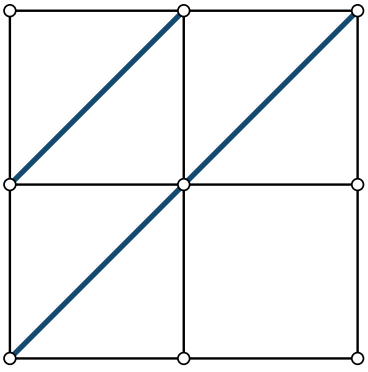
A



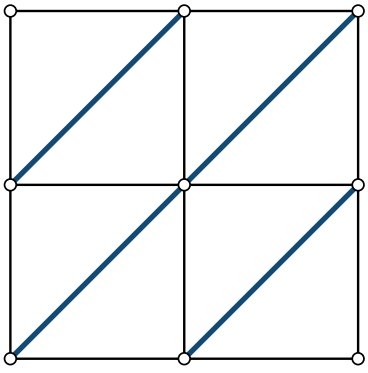
B



C

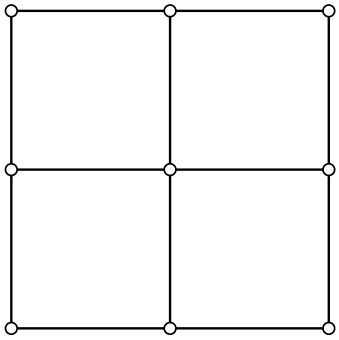


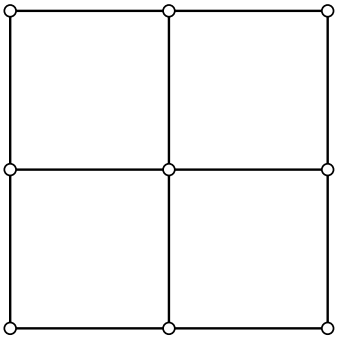
D

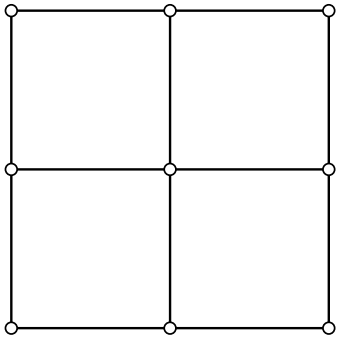


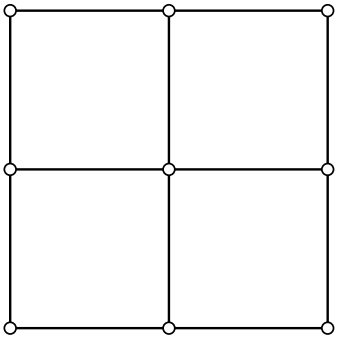
Problem 5: Find the fewest braces that make your 2-by-2 frame hold its shape. Find several designs using that many braces.

Fewest braces: _____





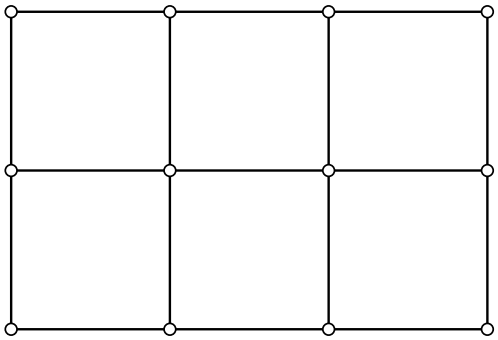


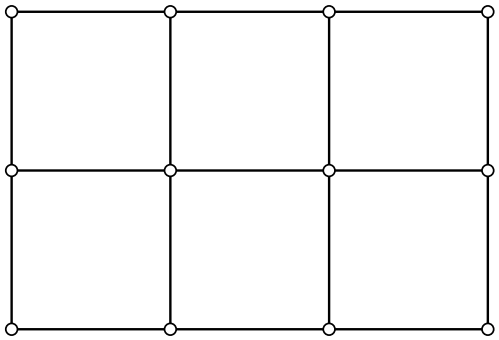


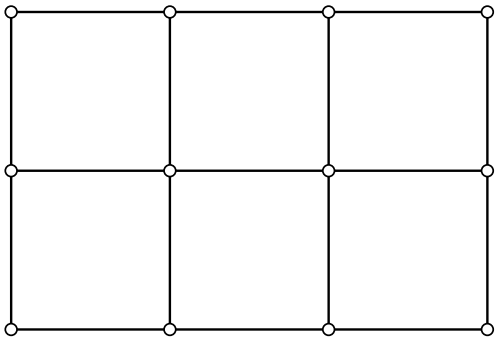
Keep the frames flat. Moving or turning the whole frame does not change its shape. Use at most one brace in each cell.

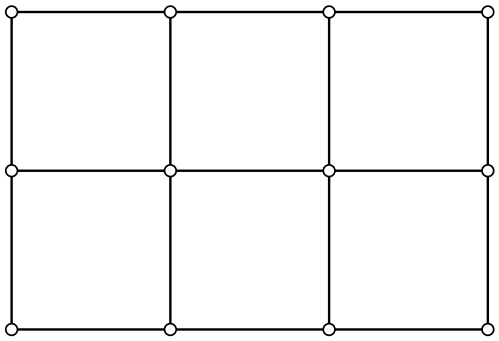
Problem 6: Find the fewest braces that make your 2-by-3 frame hold its shape. Find several designs using that many braces.

Fewest braces: _____

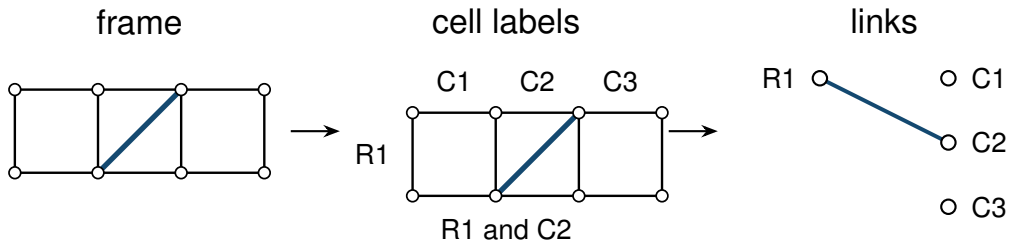




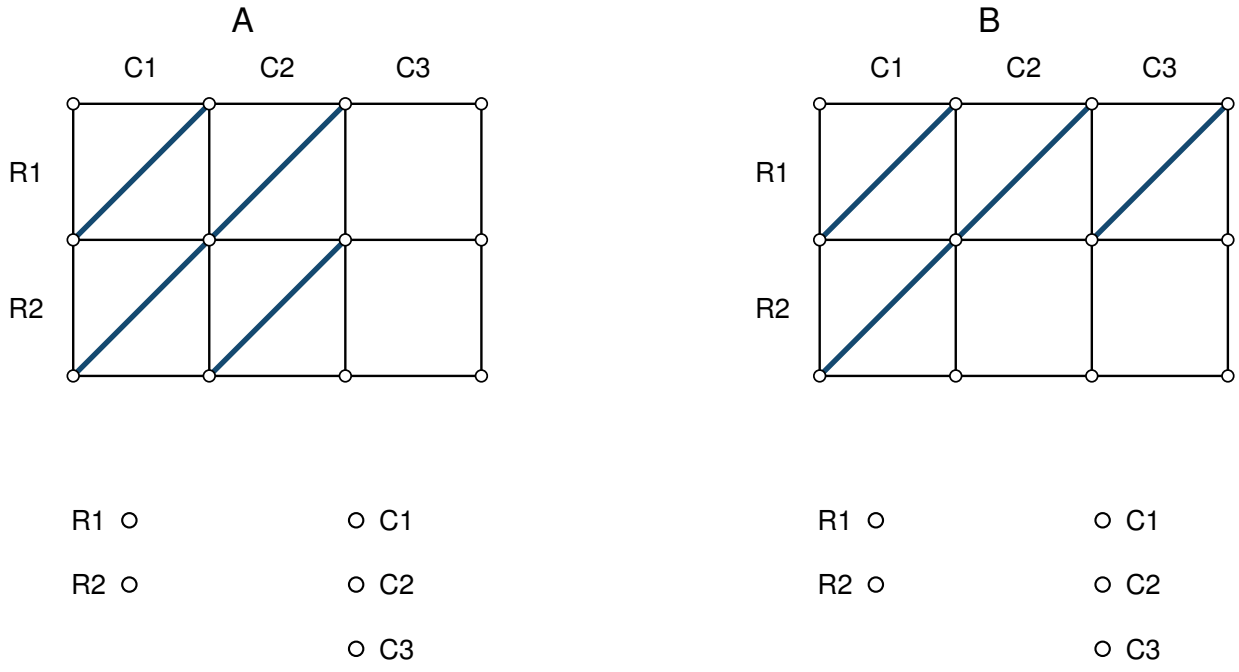




A brace becomes a link between its row dot and its column dot. Keep all the dots. Links may cross; a crossing is not a dot.

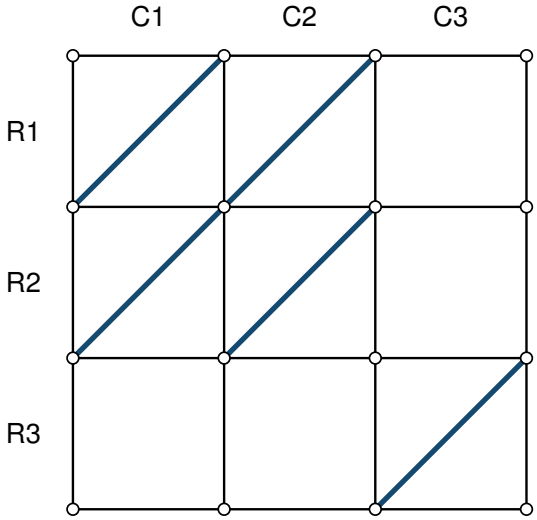


Problem 7: Make A and B on your 2-by-3 frame. Draw their links. Circle each design that holds its shape.



Problem 8: Does putting a brace in every row and every column always make a frame hold its shape? Draw A and B's links and decide.

A

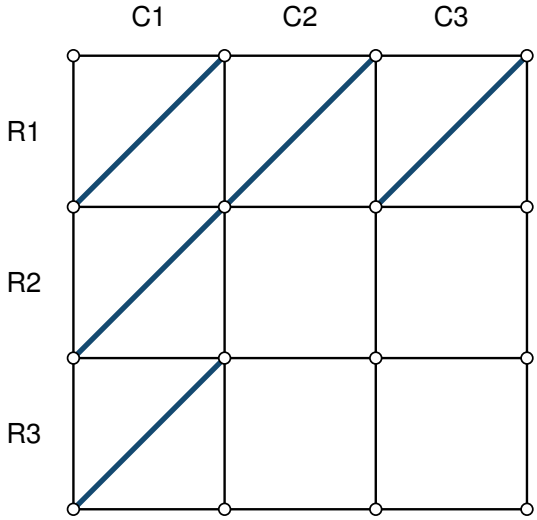


R1 ○ ○ C1

R2 ○ ○ C2

R3 ○ ○ C3

B

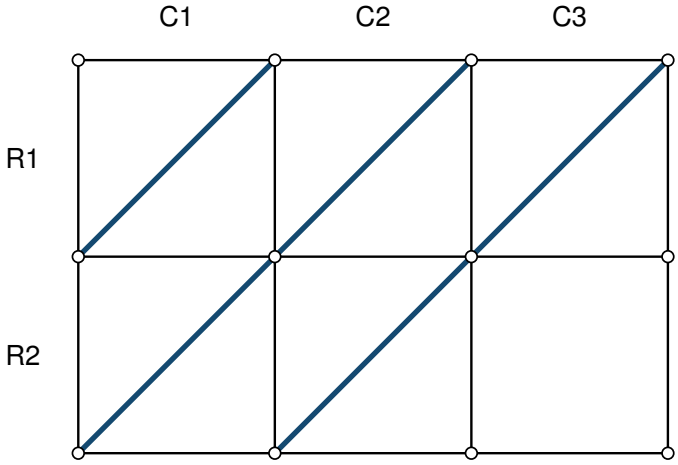


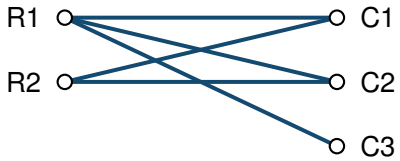
R1 ○ ○ C1

R2 ○ ○ C2

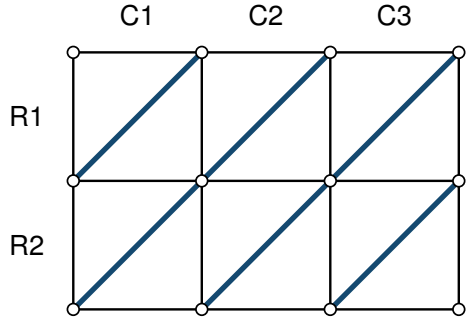
R3 ○ ○ C3

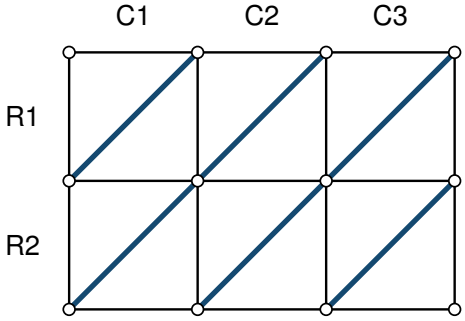
Problem 9: Find every single brace you can remove from this design while the frame still holds its shape. Replace it before testing another. Mark those braces on both pictures.

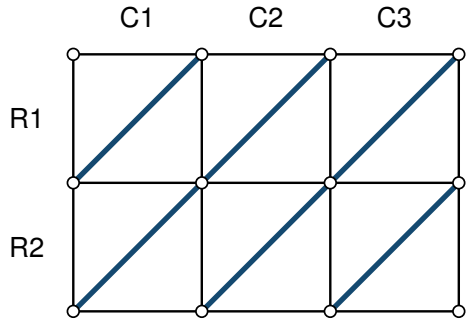


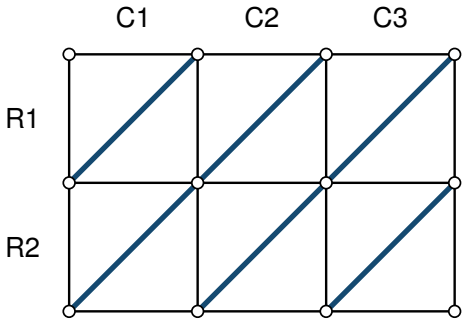


Problem 10: Starting with all six cells braced, find pairs you can remove while the frame still holds its shape, and pairs that let it change shape. Cross out the removed braces and circle each design that holds its shape.



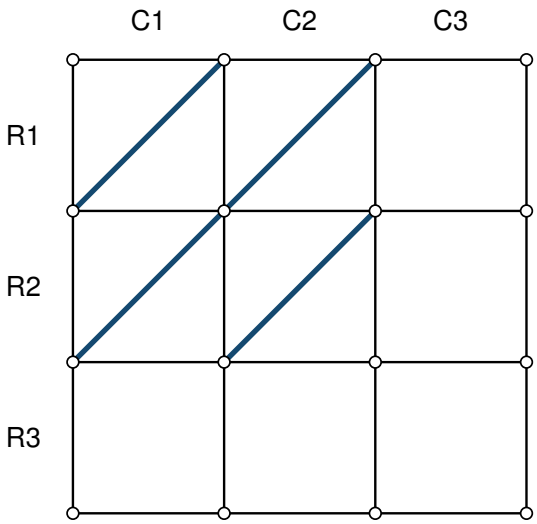






Problem 11: Can one new brace make each frame hold its shape? Mark every place that works.

A

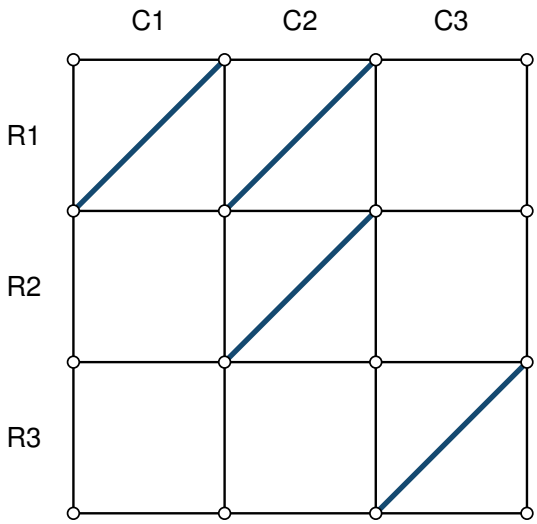


R1 ○ ○ C1

R2 ○ ○ C2

R3 ○ ○ C3

B



R1 ○ ○ C1

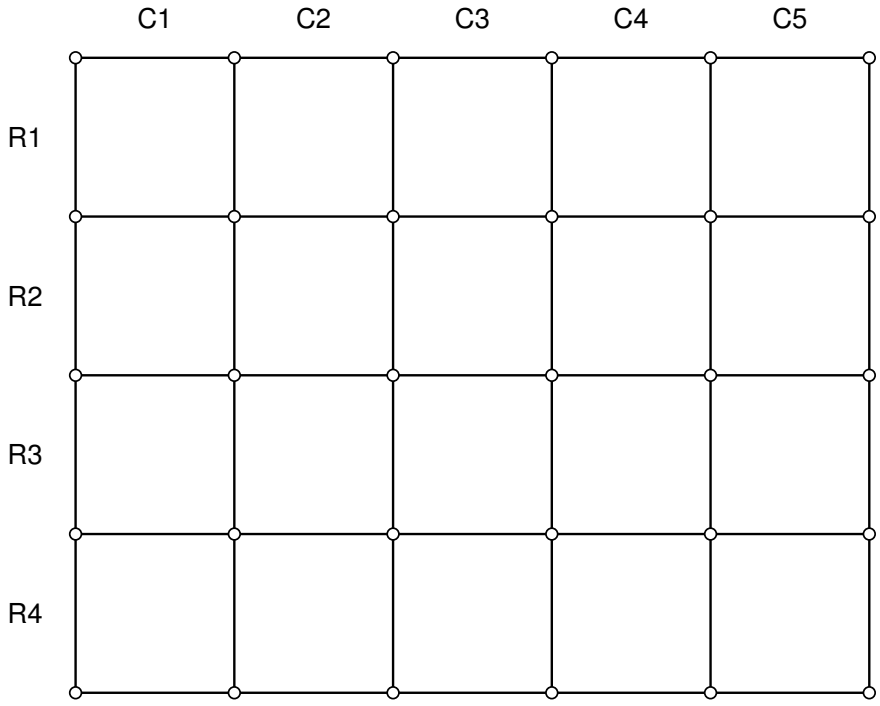
R2 ○ ○ C2

R3 ○ ○ C3

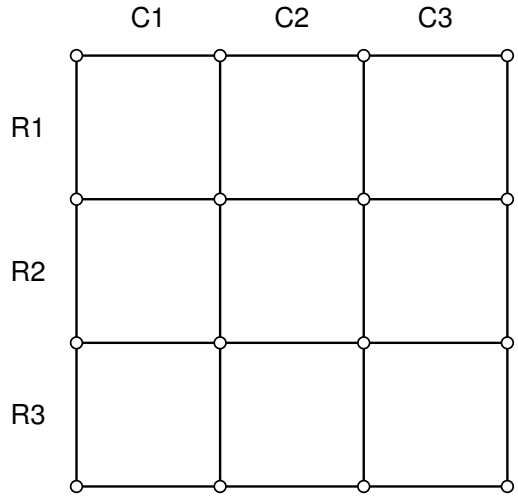
Keep the frames flat. Moving or turning the whole frame does not change its shape. Use at most one brace in each cell.

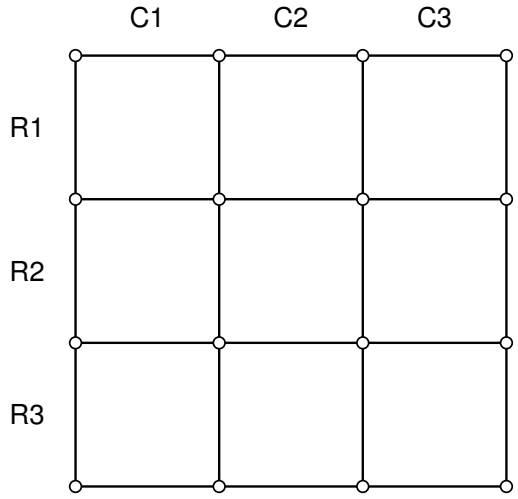
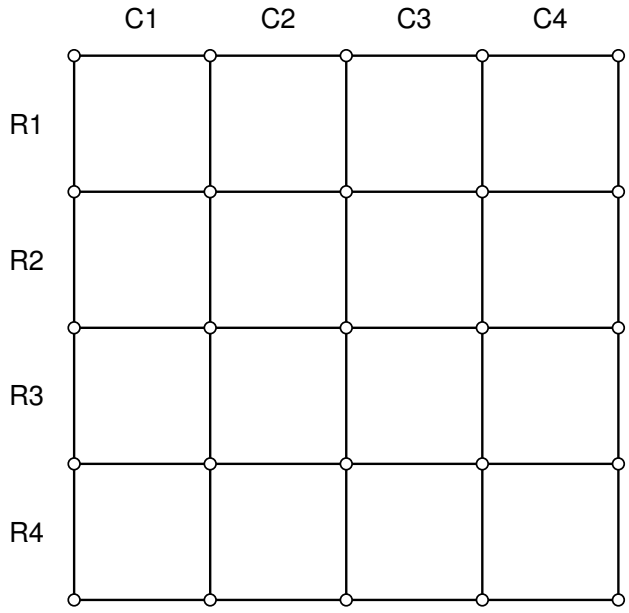
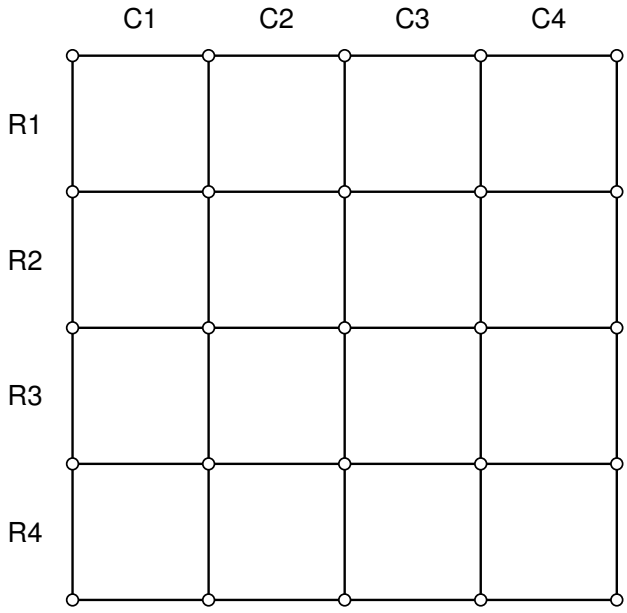
Problem 12: Find the fewest braces that make a 4-by-5 grid hold its shape. Draw a design using that many, and explain why fewer cannot work.

Fewest braces: _____



Problem 13: A 3-by-3 grid has braces in six different cells, with a brace in every row and column. Must it hold its shape? Explain.





Problem 14: A 4-by-4 grid has nine braces, with a brace in every row and column. Must it hold its shape? Explain.